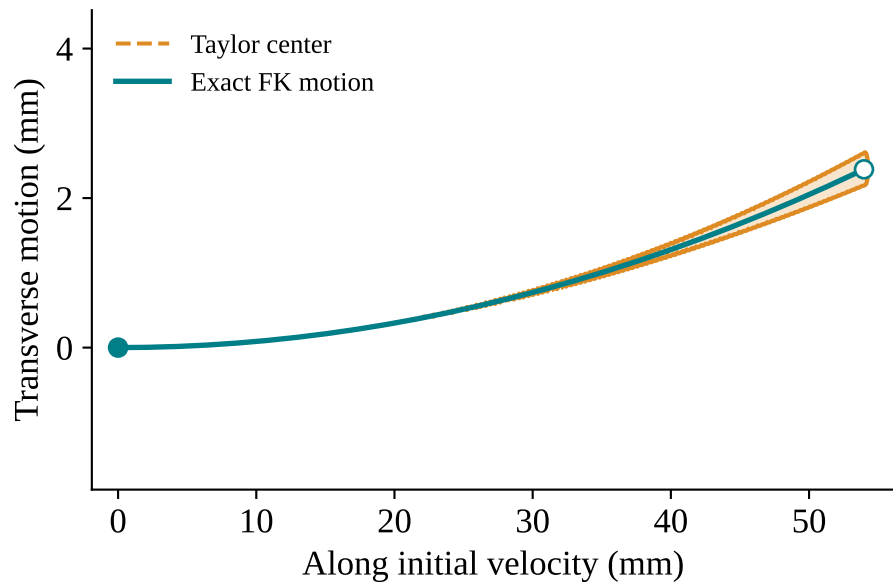


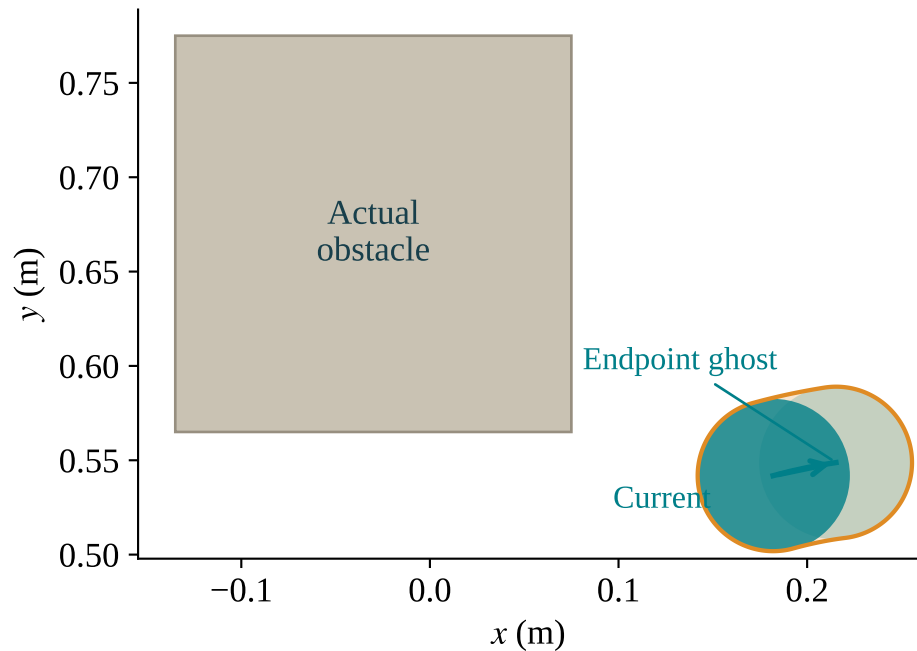
(b) Computed kinematic envelope



$$p(t) \in B(p_0 + t\dot{p}_0 + \frac{1}{2}t^2\ddot{p}_0, Ht^3/6)$$

Anchored certificate: 85.7 ms hop | endpoint error radius 0.22 mm | world margin 15 mm

(c) Entire sphere stays inside the tube



Orange: Taylor tube + sphere radius